

Project - Low Level Design

on

Healthcare Content Generator

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 2025-26

1. Introduction

The Healthcare Content Generator project focuses on developing an intelligent healthcare content generation system using Generative AI technologies. The application is designed to automatically generate high-quality healthcare-related content such as medical summaries, patient education materials, symptom explanations, and wellness recommendations using advanced AI models.

The system is built using Python and modern AI frameworks, integrating powerful Generative AI models capable of understanding medical context and producing accurate, human-like content. This allows the system to assist healthcare providers, patients, and content creators by automating the process of generating reliable and structured healthcare information.

To ensure smooth deployment and consistent performance, the application can be containerized using Docker, allowing it to run seamlessly across different environments. The system is designed to be scalable, efficient, and capable of handling multiple content generation requests simultaneously.

Overall, this project demonstrates how Generative AI can transform healthcare content creation by improving efficiency, accuracy, and accessibility of medical information.

1.1 Scope of the Document

This document describes the Low-Level Design (LLD) of the Healthcare Content Generator system. It explains the internal architecture, system components, and how Generative AI is integrated into the application.

The document covers the design of the AI-powered content generation system, including prompt processing, model interaction, and response generation workflows. It also outlines how the system ensures high-quality, context-aware healthcare content generation.

Additionally, the document explains the technologies used in the system, including AI models, APIs, and backend frameworks. It provides a clear understanding of how the system processes user inputs and generates meaningful healthcare-related outputs.

The purpose of this document is to help developers, testers, and stakeholders understand how the system is designed and how Generative AI plays a central role in the application

1.2 System Overview

The Healthcare Content Generator is an AI-powered system designed to generate healthcare-related content based on user inputs. The system allows users to enter queries such as symptoms, medical topics, or healthcare questions, and it produces structured and informative content using Generative AI models.

The core of the system is the Generative AI engine, which processes input prompts and generates relevant healthcare content. This includes explanations of diseases, treatment options, preventive measures, and patient-friendly summaries. The AI model is configured to provide accurate, safe, and easy-to-understand medical information.

The system ensures that generated content is context-aware and tailored to user needs. It supports use cases such as patient education, blog generation, healthcare documentation, and medical assistance.

The architecture is scalable, allowing multiple users to generate content simultaneously and enabling deployment across local or cloud environments.

1.3 System Design

The system design follows an AI-driven architecture where Generative AI is the core component.

The process begins with the user providing a healthcare-related query. The backend system processes this input and prepares it for the Generative AI model.

The backend, built using Python and Flask, handles input validation, prompt structuring, and communication with the AI model. The processed prompt is sent to the Generative AI engine.

The AI model generates context-aware healthcare content, which is then refined and formatted before being returned to the user.

This design ensures efficient content generation, scalability, and seamless interaction between users and the AI system.

2. Data Design

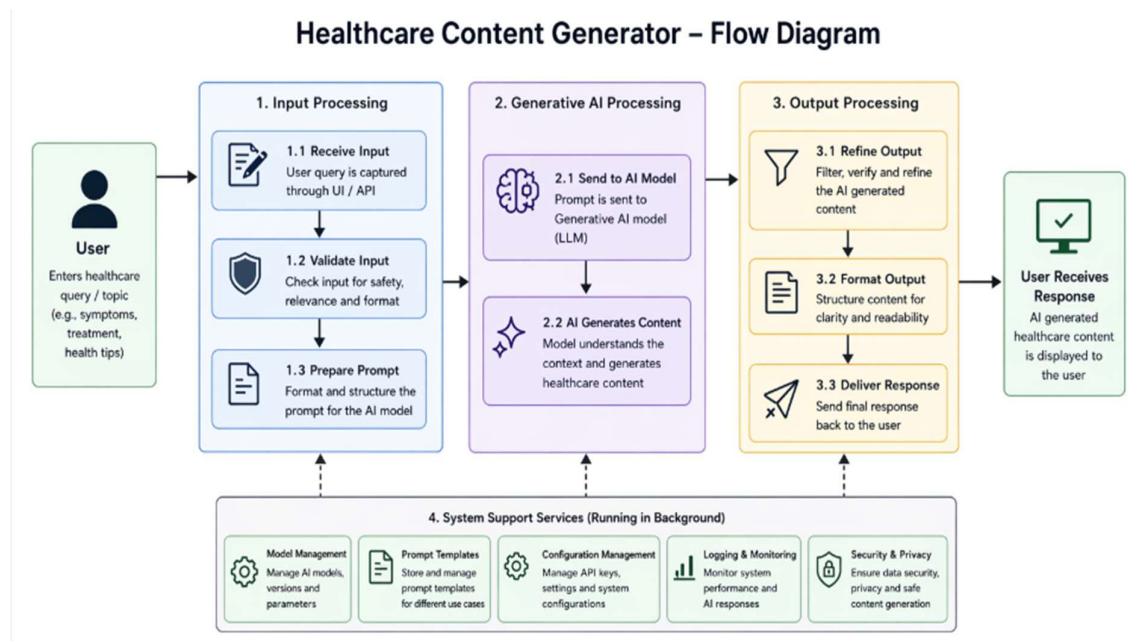
The Data Design defines how input data and generated content are managed within the system.

The system processes user inputs such as healthcare queries, symptoms, and topics. These inputs are structured before being sent to the AI model.

The output consists of AI-generated healthcare content, including explanations, summaries, and recommendations.

The system also manages prompt templates, response formatting rules, and configuration data such as API keys and model parameters.

This ensures smooth data flow and consistent output generation.



3. Interfaces

4. State and Session Management

The system manages user sessions to handle requests and responses efficiently.

Each request is processed independently, ensuring no conflicts between users.

Advanced implementations may maintain conversational context to improve response relevance.

This ensures smooth and accurate interaction between users and the AI system.

5. Functional Requirements

5.1 Input Processing

1. User Query Input:
The system must allow users to input healthcare-related queries.
2. Prompt Formatting:
The system must structure input for AI processing.
3. Input Validation:
The system must validate inputs for correctness and relevance.

5.2 Generative AI Processing

1. AI Content Generation:
The system must generate healthcare content using AI.
2. Context Understanding:
The AI must understand input context accurately.
3. Response Optimization:
The system must refine outputs for clarity and readability.

5.3 Output Delivery

1. Response Generation:
The system must deliver AI-generated content to users.
2. Structured Output:
The output must be clear and well-organized.
3. Real-Time Processing:
The system must generate responses quickly.

5.4 System Management

1. Model Integration:
The system must integrate with AI models.
2. Configuration Management:
The system must manage model parameters effectively.
3. Monitoring:
The system should monitor performance and output quality.

5.5 AI Workflow

A. Input Stage

1. User provides query
2. System processes input

B. AI Processing Stage

1. Input sent to AI model
2. Model generates content

C. Output Stage

1. Content is refined
 2. Response delivered to user
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6. Non-Functional Requirements

6.1 Security Aspects

The system must ensure secure handling of user data and restrict unauthorized access.

6.2 Performance Aspects

The system must generate responses efficiently with minimal delay.

6.3 Scalability

The system must support multiple concurrent users and scale as demand increases.

6.4 Reliability and Availability

The system must ensure consistent performance and minimal downtime.

6.5 Maintainability

The system must be easy to update and manage without affecting performance.

7. References

1. OpenAI Documentation (2024)
2. Hugging Face Documentation (2024)
3. Python Documentation (2024)
4. Generative AI Research Papers (2023)
5. Flask Documentation (2024)